

Monte Carlo Simulation for Multistate Credit Portfolios

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1 ABSTRACT

A typical loan in a credit portfolio can take a range of credit states (this paper assumes S&P ratings, however CMS ratings can be implemented with ease), and migrations between these states are commonplace, bringing with it a large cost penalty to the portfolio value. A typical binary model does not capture these costs, which in turn results in dangerously low predictions for *Expected Shortfall* (also referred to as CVaR) and *Value at Risk* (shortened to VaR). The aim of this paper is to show how this issue can be remedied using a simple Monte Carlo simulation, and the tests involved in verifying our results.

2 PORTFOLIO IN USE

In order to understand the behaviour of the model, we must first understand the portfolio we are using. Here, we use a synthetic portfolio consisting of 50,000 loans, with a constant ρ^2 value of 0.2 for each loan. The overall EAD of the portfolio is £13.361b, with this figure distributed log-normally across our assets. From here, we assign three attributes to each loan: [1] a credit rating (AAA,..., default), [2] a sector (Corporate, FI and CRE) and [3] a region (US or UK, split 70%/30%). The sector and credit rating proportions are as follows:

Table 1: Portfolio Weighting by Credit Rating

Credit Rating	% of Portfolio
AAA,CCC	5%
AA,B	10%
A,BB	20%
BBB	30%

Table 2: Portfolio Weighting by Sector

Sector	% of Portfolio
FI	20%
CRE	30%
Corporate	30%

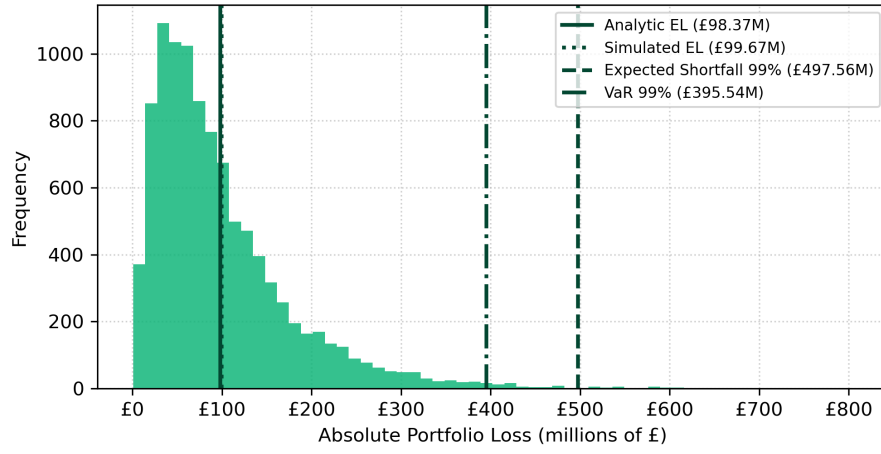
3 INITIAL PREPARATION

The binary default model created by Oliver Gregory (2025) served as the groundwork for the development of this model. The default mechanism was defined as follows:

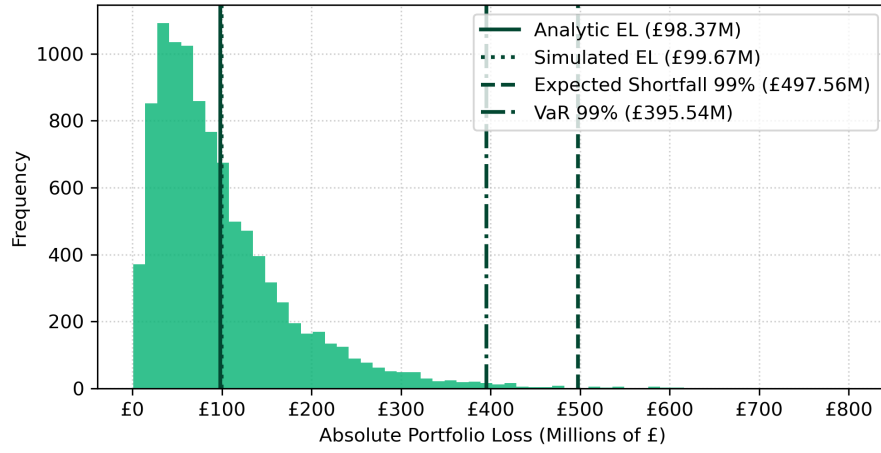
$$\mathbb{P}(X_i \leq d_i) = PD_i \implies d_i = \Phi^{-1}(PD_i)$$

where X_i is the latent asset return on obligor i and PD_i denotes the probability of default (over one year) on said obligor. Φ^{-1} denotes the inverse standard normal CDF.

Our first task was to alter the provided code to implement the framework involved with taking in a *transition matrix*. We can not immediately test the newly updated code as we have no basis for comparison, so instead we verify the logic of the code under a boundary condition. This boundary condition effectively forces the probability of migrations between non default states to be zero, which has the effect of 'collapsing' the multistate assumption to a binary one. Unsurprisingly, we observe identical results for the loss distribution, tail quantiles and risk metrics, as shown in figure 1.



(a) Binary Model



(b) Multistate Boundary Condition

Figure 1: Comparison of Binary and Multistate (Boundary) Results

4 REMOVING BOUNDARY CONDITIONS

With the framework implemented, we are nearly ready to run some tests. However, we need to make a minor modification to the existing code. *Residual Maturity* is another data point included on a typical loan book, and is simply defined to be the time to maturity as measured from the current moment. Our existing portfolio did not take this into account, which detracts from the practicality of the model. To fix this, we implemented a residual maturity vector $\vec{\mu} \in \mathbb{R}^M$, defined as:

$$\vec{\mu} = (\mu_1, \dots, \mu_M)^T, \quad \mu_i \sim U[1, 10] \quad \forall i \in \{1, \dots, M\}$$

where μ_i denotes the residual maturity on obligor i , uniformly distributed between 1 and 10 and M denotes the number of obligors.

In order to understand the significance of residual maturities, we must also understand the notion of a transition matrix. A transition matrix, denoted T , gives the probability of migrating between any two credit states for a given asset. In order to calculate the probability of migration from state α to state β for obligor i over our 1 year simulation horizon, we must take into account our residual maturities:

$$\mathbb{P}(\alpha \rightarrow \beta) = 1 - (1 - T_{\alpha\beta})^{\mu_i}$$

Summing over all possible transitions, we can calculate a lifetime expected loss vector, denoted $\vec{\ell}_i$ on our obligor:

$$\begin{aligned} \vec{\ell}_i &= EAD_i LGD_i \sum_{s \in \{\text{states}\}} \mathbb{P}(\alpha_i \rightarrow s) \vec{e}_s \\ &= EAD_i LGD_i \sum_{s \in \{\text{states}\}} [1 - (1 - T_{\alpha_i s})^{\mu_i}] \vec{e}_s \end{aligned}$$

where α_i is the current state of the loan, and $\{\text{states}\}$ is the set of possible states that the loan can migrate to (for our S&P ratings, this will be AAA, AA, ..., CCC, default). T_{ij} represents the probability of transitioning from state i to j , found by looking at the transition matrix. Vectors $\vec{e}_s$ form an orthogonal basis, i.e. $\vec{e}_i \cdot \vec{e}_j = \delta_{ij}$.

One major assumption made in this model is that this is a constant attribute among all assets in the loan book; in reality this is far from true, with each asset having a unique transition matrix associated to it.

From/To	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	0.911209	0.079992	0.006999	0.001000	0.000500	0.000100	0.000100	0.000100
AA	0.006999	0.910118	0.074685	0.005999	0.001000	0.000700	0.000200	0.000300
A	0.001000	0.023395	0.915217	0.050790	0.006099	0.002599	0.000100	0.000800
BBB	0.000200	0.002998	0.056472	0.879359	0.047476	0.010495	0.000999	0.002000
BB	0.000100	0.001100	0.005499	0.077692	0.817617	0.079492	0.008499	0.010000
B	0.000000	0.000500	0.002500	0.004500	0.070000	0.835000	0.037500	0.050000
CCC	0.000000	0.000100	0.001000	0.003000	0.025900	0.120000	0.650000	0.200000
Default	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.000000

Figure 2: Transition Matrix Used in Analysis

Figure 2 gives the transition matrix we fed into the model. Adapting the model to CMS ratings should be fairly trivial as the calculations would be identical. One would simply need to feed an $\mathbb{R}^{20 \times 20}$ matrix as opposed to the $\mathbb{R}^{8 \times 8}$ matrix above, along with a new PD mapping vector. It should be noted that the probability of a downward migration is significantly higher than an upwards migration; as an example, it can be observed that AAA→BBB has a 0.1% probability, while BBB→AAA has a 0.02% probability. This will have some bearing on the results we obtain.

Taking these two facts into account, one can hypothesise that our new loss distribution will result in far higher losses. Below, we observe exactly that:

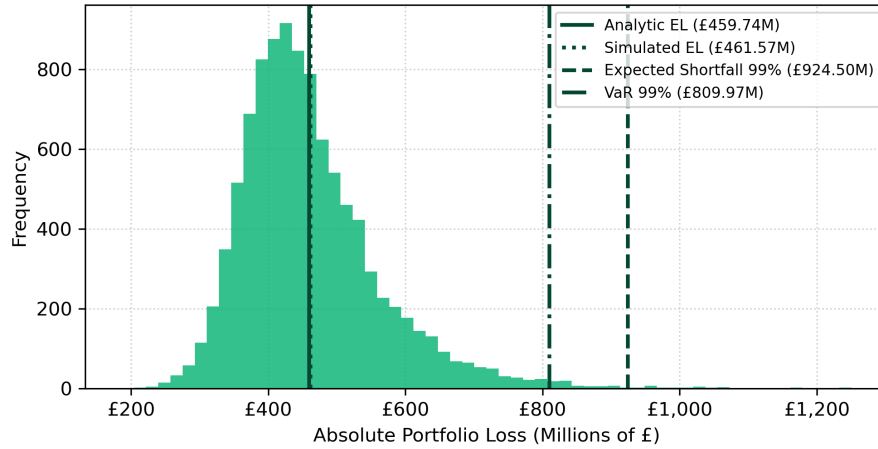


Figure 3: Multistate + Residual Maturity Loss Distribution

The reasoning behind this shape is fairly intuitive; by implementing a transition matrix one has two compounding effects that vastly increase the losses. Assuming a scenario in which we have a perfectly symmetric transition matrix, we should still expect to see higher losses, as downward migrations destroy significantly more portfolio value than the corresponding upward migration will create. On top of this, our threshold matrix is **not** symmetric, but rather weighted towards downward migrations, further ballooning our losses.

5 TESTS

The results we have obtained are reassuring, but don't serve as definitive proof of validity. In this section, we briefly describe a series of tests to provide a clearer picture on the behaviour of the model.

5.1 LOAN LEVEL CVAR

The first test aimed to show that our loan level *CVaR* at 99% (that is, the expected losses assuming we are past the 99th tail quantile) summed to the CVaR on a portfolio level. In order to calculate this, we slightly modified our code to be able to: [1] calculate the VaR and CVaR at 99% on a portfolio level on the first pass (as we did before), and [2] run the model a second time to find the CVaR contribution of each loan in the 1% worst case scenarios. Below are the results of the test:

Table 3: Verification of Portfolio-Level CVaR and Cumulative Sum

Metric	Value (£M)
Running Cumulative Sum (99%)	924.4952
Portfolio-Level CVaR (99%)	924.4952
Absolute Difference	2.27×10^{-12}

As visible, the cumulative sum and portfolio level CVaRs match up very closely, providing further verification on our model.

5.2 CONVERGENCE TEST

Another test we ran was a convergence test. If the model is working as intended, we should expect to see the simulated EL value to converge to the analytic EL value, which was calculated as follows:

$$EL_{\text{analytic,multistate}} = \sum_{i=1}^M \sum_{s \in \{\text{states}\}} \mathbb{P}(\alpha_i \rightarrow s) \times \text{Loss}_{i,s}$$

where $\mathbb{P}(\alpha_i \rightarrow s)$ is defined as before, and $\text{Loss}_{i,s}$ represents the loss associated with that specific destination state.

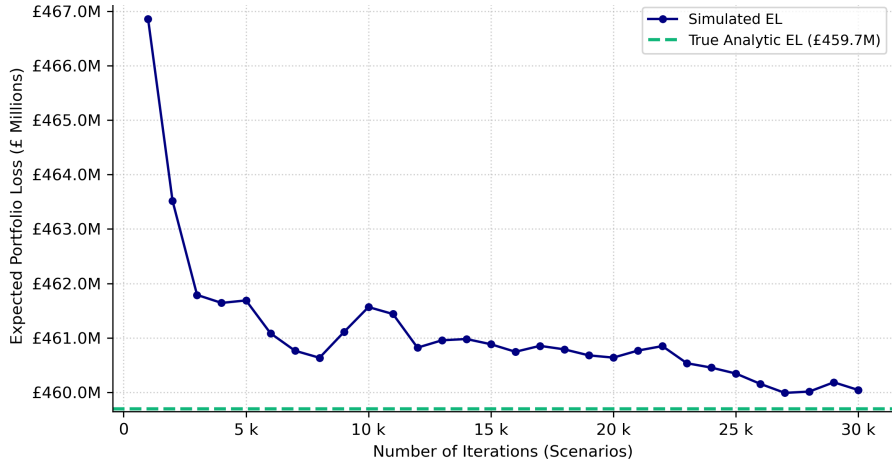


Figure 4: Convergence of the model up to 30k iterations

We can clearly see a strong convergence to the analytic result, with a % difference of about 0.065% at 30,000 iterations. It should however be noted that this result relies exclusively on the accuracy of the analytic engine. In our case, we have opted for the simple calculation that we put forth on page 5, however a more precise model proposed by Mikhail Voropaev (2011) is guaranteed to act as a better benchmark. The accuracy of the analytic result can be tested using more iterations (e.g. 1,000,000), however that is beyond the scope of this paper.

5.3 SECTOR LEVEL CVAR

The third and final test we ran was to test the model under highly non granular portfolio conditions, with varying ρ^2 values. Here, we reverted to the binary model, as this allows for greater control over the input variables, and removes a significant chunk of Monte Carlo noise. Our aim in these examples is to observe how the asset level *CVaR rate* (i.e. tail loss per £1 of exposure in a specific asset, calculated by Asset CVaR $\div$ Asset EAD) changes under varying conditions. In all examples, it can be assumed that the following is constant for each loan: [1] EAD per loan = £267,220, [2] Portfolio EAD = £13.361b [3] LGD = 0.3, [4] credit rating = BBB, [5] number of loans = 50,000. As done in all other examples, we run the simulation for 10,000 iterations.

For our first example, we compared the asset specific CVaR rates for two portfolios, both with $\rho^2 = 0.2$ and an exposure to UK CRE loans of 50% and 90% respectively. Intuition tells us that the CVaR rates for UK CRE assets should be higher in the 90% portfolio when compared to the 50%. This is because we have a significant exposure to two systematic factors, those being UK specific and CRE specific shocks. In our tail scenarios, both UK markets and CRE sectors have experienced large losses, which compounds to larger losses in our portfolio. To keep it short: don't put all your eggs in one basket. Below are the results from that test:

Table 4: $\rho^2 = 0.2$, UK CRE = 90%

Region / Sector	CVaR (£M)	EAD (£M)	CVaR Rate (%)
UK	119.12	12,556	0.9487%
Corporate	2.13	259	0.8237%
CRE	115.14	12,024	0.9575%
FI	1.85	273	0.6798%
US	5.41	804	0.6727%
Corporate	1.74	275	0.6324%
CRE	2.16	266	0.8099%
FI	1.51	263	0.5759%
Grand Total	124.53	13,361	0.9321%

Table 5: $\rho^2 = 0.2$, UK CRE = 50%

Region / Sector	CVaR (£M)	EAD (£M)	CVaR Rate (%)
UK	82.88	9,388	0.8827%
Corporate	11.90	1,359	0.8756%
CRE	61.44	6,680	0.9197%
FI	9.54	1,349	0.7070%
US	30.59	3,970	0.7701%
Corporate	10.22	1,325	0.7708%
CRE	11.40	1,303	0.8745%
FI	8.97	1,342	0.6680%
Grand Total	113.46	13,361	0.8492%

Observe the CVaR rates for UK CRE assets. We can see a slight increase in the rate, with a % increase of approximately 4%, which aligns with our predictions. Note also how our portfolio CVaR has increased by over £11m; these results strongly imply that our portfolio has become riskier, which is consistent with the fact that the portfolio is significantly less diversified.

Our second test aimed to observe the effects of increasing the ρ^2 sensitivity on our 50% UK CRE portfolio. In doing so, we are increasing the loan sensitivity to systematic shock, which should hopefully result in higher CVaR rates across the board.

Table 6: $\rho^2 = 0.6$, UK CRE = 50%

Region / Sector	CVaR (£M)	EAD (£M)	CVaR Rate (%)
UK	301.00	9,371	3.2120%
Corporate	42.30	1,316	3.2139%
CRE	225.45	6,680	3.3748%
FI	33.26	1,347	2.4189%
US	112.42	3,989	2.8179%
Corporate	38.98	1,323	2.9469%
CRE	41.32	1,321	3.1289%
FI	32.12	1,346	2.3860%
Grand Total	413.43	13,361	3.0943%

Again, we see exactly what we predicted: dramatically higher CVaR rates on all asset types. The explanation for this follows. Recall the definition of asset return ξ_i on obligor i :

$$\xi_i = \sqrt{\rho^2} \vec{\beta}_i \cdot \vec{\eta} + \sqrt{1 - \rho^2} \epsilon_i$$

where $\vec{\beta}_i$ gives asset specific weightings to systematic factors, $\vec{\eta} \in \mathbb{R}^N$ is a systematic factor vector (in our case, $N = 5$ as we have 5 systematic factors) and ϵ_i is the idiosyncratic risk.

When we increase ρ^2 , we are effectively shifting shock dependence from idiosyncratic to systematic, making our portfolio far more vulnerable to market shifts. In the case where $\rho^2 \rightarrow 1$, we have a portfolio entirely dependent on systematic risk, with asset return given by $\xi_i = \beta_i \cdot \vec{\eta}$. The results in tables 5 and 6 show the transition from asset return mainly dependent on idiosyncratic risk to systematic risk.

Our third and final example focused on the extremely niche case of a portfolio with $\rho^2 = 0.0$. This can be thought of as a portfolio whose investments are perfectly insulated from outside market movements (e.g. investments on businesses that exist on desert islands). Here, we reverted to the 50%/90% comparison from earlier. Using the same logic as we used earlier, with $\rho^2 = 0.0$, one obtains an asset return given by $\xi_i = \epsilon_i$. This effectively implies **no** dependence on the asset type, that is, the portfolio shouldn't care how it is structured. Indeed, we observe exactly that:

Table 7: Portfolio Risk Metrics Comparison ($\rho^2 = 0.0$)

(a) $\rho^2 = 0.0$, UK CRE = 50%

Region / Sector	CVaR (£M)	EAD (£M)	CVaR Rate (%)
UK	7.13	9,374	0.0760%
Corporate	1.05	1,341	0.0781%
CRE	5.08	6,680	0.0761%
FI	1.00	1,352	0.0737%
US	3.09	3,986	0.0776%
Corporate	1.07	1,332	0.0801%
CRE	1.02	1,324	0.0770%
FI	1.01	1,331	0.0755%
Grand Total	10.22	13,361	0.0765%

(b) $\rho^2 = 0.0$, UK CRE = 90%

Region / Sector	CVaR (£M)	EAD (£M)	CVaR Rate (%)
UK	9.59	12,567	0.0763%
Corporate	0.21	268	0.0796%
CRE	9.17	12,025	0.0762%
FI	0.21	273	0.0762%
US	0.63	794	0.0794%
Corporate	0.21	259	0.0819%
CRE	0.21	265	0.0779%
FI	0.21	271	0.0785%
Grand Total	10.22	13,361	0.0765%

While we do see some small discrepancies among the CVaR Rate, this likely arises due to Monte Carlo noise. Indeed, when looking at our total CVaR rates, we see the same figure, providing a verification of our hypothesis.

6 CONCLUSION

The results and tests discussed in this paper are all well documented and hence, it should come as no surprise when we say that this model is crude in comparison to more sophisticated analytic engines proposed by Voropaev (2011) and Buet-Golfouse (2015). These models are extremely versatile due to their drastically faster compute times compared to a standard Monte Carlo approach. However, the aim of this paper is not to propose any groundbreaking concepts, but rather to give an overview on the process of developing and testing such a model. In doing so, the reader should have hopefully gained a better understanding on the limitations of the model, and any areas where assumptions affect the quality of the results obtained.

7 REFERENCES

Gregory, O (2026). Estimating a credit portfolio loss distribution using monte-carlo simulation. *Available at:*

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Buet-Golfouse, F. (2015). The application of Hermite polynomials to risk allocation. *Risk* 18(3), 77-110